

Rapid Approximate Calculation of Numbers of Quantum States $W(E, J)$ in the Phase Space Theory of Unimolecular Bond Fission Reactions

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A facile and rapid approximation for counting numbers of quantum states $W(E, J)$ is described for application in the phase space theory of unimolecular bond fission reactions. The method accounts for J -conservation. It is based on an empirical interpolation between analytical expressions of $W(E, J)$ in the low J - and high J -limits, and it is checked by extensive accurate state counting for sets of representative model molecules.

1. Introduction

The counting of numbers of quantum states forms an essential part of statistical theories of rate processes such as transition state theory, quasi-equilibrium theory, phase space theory, RRKM theory, statistical adiabatic channel models, and other statistical treatments. For this reason, efficient counting algorithms for numbers of quantum states $W(E, J)$ are required (E = total energy, J = total angular momentum of the system). If the system is composed only of harmonic oscillators, the Beyer-Swinehart algorithm^[1,2] is particularly useful, because it requires a minimum amount of computer time such that very large systems can also be handled. The success of this method is due to the very economical numerical execution of the repetitive convolution of the number of states $W_{s-1}(E)$ of a subset of $s-1$ harmonic oscillators with the density of states $\varrho(E)$ of an additional oscillator via the relation

$$W_s(E) = \int_0^E W_{s-1}(E-E') \varrho(E') dE'. \quad (1.1)$$

Eq. (1.1) in integral form symbolizes the discrete convolution operation.

The Beyer-Swinehart algorithm can be extended to include anharmonic oscillators and free or hindered rotors^[2]. However, if accurate counting is required, the efficiency of the algorithm may be strongly reduced. One way out of this dilemma^[3] is the separate determination of the combined number of states of all anharmonic oscillators, free or hindered rotors and the subsequent repetitive convolution with harmonic oscillators via Eq. (1.1). In this way, the advantages of the Beyer-Swinehart algorithm are conserved for that part of the system which is characterized by equidistant energy levels. The other part, which does not allow for the application of the algorithm, is treated separately and enters via the starting array of the counting algorithm^[1,3].

Convolutions of numbers and densities of states of harmonic oscillators and sets of coupled rotors are particularly important in phase space theory (i.e., in loose transition state theory). Activated complex states here are approximated by vibrational levels of separated species and rotational levels of the motion of the species relative to each other. Activated complexes are located at the top of centrifugal barriers which, for simple bond fissions, are situated at large distances between two separating species such that the vibrational levels are close to those at infinite separation. The complication lies in the coupling of the fragment rotations expressed by the conservation of the total angular momentum J . Provided that the number of states $W_i(E, J)$ of the rotors is known, the convolution with harmonic oscillator states via Eq. (1.1) leads to the total number $W(E, J)$ of activated complex states which, e.g., enters the expression for specific rate constants $k(E, J)$ of a unimolecular reaction

$$k(E, J) = \frac{W(E, J)}{h \varrho(E, J)}. \quad (1.2)$$

If the convolution is done with the Beyer-Swinehart algorithm, the most efficient access to $k(E, J)$ is obtained.

It is the aim of the present article to review and extend expressions for numbers of states $W_i(E, J)$ of coupled rotors in phase space theory such that rapid computations of $k(E, J)$ become possible. Phase space theory (or loose transition state theory) is known to overestimate rate constants because it assumes an isotropic interaction potential between the separating species (or the approaching species in bimolecular reaction events). Although it neglects "rigidity" of the activated complexes, which is introduced by the anisotropy of the potential, it provides a useful first approximation, e.g., for unimolecular fragmentation rates. The present expressions for $W_i(E, J)$ and their convolution with harmonic oscillator contributions via Eq. (1.1) apply to this approach. However, they also apply to the next level of approximation, the "adiabatic channel modified phase space theory" of Refs. [4] and [5], in which the rotational constants of the rotors are increased by energy independent factors accounting for an "average rigidity" of the potential

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surface. On a further level of approximation, accurate adiabatic channel threshold energies would have to be determined and the corresponding $W_l(E, J)$ calculated, before the Beyer-Swinehart convolution of Eq. (1.1) is applied. The results then should be compared with phase space theory in order to appreciate the consequences of rigidity of the potential energy surface.

2. General Relationships for Numbers of Coupled Rotational States $W_l(E, J)$

We consider two nonlinear species with angular momentum vectors j_1 and j_2 and an angular momentum vector l of relative orbital motion. These angular momenta can be combined in a number of ways to form a total angular momentum J . Neglecting the $2J + 1$ orientations of J in space, which introduces a cancelling factor $2J + 1$ both in $W(E, J)$ and $q(E, J)$ in Eq. (1.2), the number of ways of combining j_1, j_2 , and l to form J defines the number of rotational states $W_l(E, J)$. An accurate general determination of $W_l(E, J)$ is possible only by numerical techniques, see e.g. Refs. [6–9]. In the following, we review the special cases, for which explicit expressions are available, before we proceed to general empirical but approximate formulae. We follow the treatment of appendix C2 in Ref. [10]. Cases with varying complexity of the two species have to be distinguished. For simplicity, we only consider atomic, linear and spherical top species. The expressions can easily be extended to include symmetric and asymmetric tops.

2.1. Atom + Linear Molecule Systems

At first we consider a system in which one of the separated species is an atom, the other a linear molecule with the rotational constant B_∞ (in energy units). We assume that the rotating system, for a total angular momentum J , has a well defined centrifugal barrier energy of $E_0(J)$ such that the energy

$$E^* = E - E_0(J) \quad (2.1)$$

can be distributed over the rotations of the separated species and the relative translational motion. The number of accessible quantum states then is given by

$$W_l(E, J) = \sum_j \sum_l 1 \quad (2.2)$$

where the quantum numbers j (for rotation of the linear molecule) and l (for orbital motion) are constrained by the triangular relation

$$|J - j| \leq l \leq J + j, \quad (2.3)$$

and the rotational energy $B_\infty j(j + 1)$ of the linear molecule must be in the allowed range

$$0 \leq B_\infty j(j + 1) \leq E^*. \quad (2.4)$$

Denoting the largest possible value of j by $j_{\max}$, such that

$$B_\infty j_{\max}(j_{\max} + 1) \leq E^*, \quad (2.5)$$

$W_l(E, J)$ can be calculated explicitly. One obtains [10]

$$W_l(E, J) = \sum_{j=0}^J \sum_{l=J-j}^{J+j} 1 + \sum_{j=J+1}^{j_{\max}} \sum_{l=j-J}^{j+J} 1 \quad (2.6)$$

$$= (2J + 1)(j_{\max} + 1) - J(J + 1) \quad (2.7)$$

for $0 \leq J \leq j_{\max}$, and

$$W_l(E, J) = \sum_{j=0}^{j_{\max}} \sum_{l=J-j}^{J+j} 1 \quad (2.8)$$

$$= (j_{\max} + 1)^2 \quad (2.9)$$

for $j_{\max} \leq J$. At small J and large E^* , $W_l(E, J)$ is approximated by

$$W_l(E, J) \approx (2J + 1)(j_{\max} + 1) \quad (2.10)$$

whereas, at large J , $W_l(E, J)$ according to Eq. (2.9) becomes independent of J (except for the dependence of the distributable energy E^* on the centrifugal threshold energy $E_0(J)$, see Eq. (2.1)). The goal of the present article is the derivation of a general interpolation between the easily accessible expressions for $W_l(E, J = 0)$ and $W_l(E, \text{large } J)$. For the atom + linear molecule case, this general interpolation can be checked against the accurate expression of Eqs. (2.6)–(2.9).

2.2. Atom + Spherical Top Systems

Similar to atom + linear molecule systems, one obtains explicit expressions for $W_l(E, J)$ in atom + spherical top systems. Taking into account the additional degeneracy of the top, one has

$$W_l(E, J) = \sum_j (2j + 1) \sum_l 1 \quad (2.11)$$

with the same constraints (2.3) and (2.4) as in section 2.1. Evaluating the sums of Eq. (2.11) leads to [10]

$$W_l(E, J) = (2J + 1)(j_{\max} + 1)^2 - (2J + 1)(J + 1)J/3 \quad (2.12)$$

for $0 \leq J \leq j_{\max}$, and

$$W_l(E, J) = (j_{\max} + 1)(2j_{\max}/3 + 1)(2j_{\max} + 1) \quad (2.13)$$

for $j_{\max} \leq J$. For small J and large E^* , Eq. (2.12) is approximated by

$$W_l(E, J) \approx (2J + 1)(j_{\max} + 1)^2 \quad (2.14)$$

whereas, for large J , $W_l(E, J)$ according to Eq. (2.13) becomes independent of J . Eqs. (2.12) and (2.13) again can be used as a check for a general interpolation between small J - and large J -expressions of $W_l(E, J)$.

2.3. Linear + Linear Molecule, Linear Molecule + Spherical Top, and Spherical Top + Spherical Top Systems

For systems in which neither of the two separated species is an atom, the angular momenta of the two species j_1 and j_2 couple to form an angular momentum j which in turn couples with the orbital angular momentum l to form the total angular momentum J . The constraints

$$0 \leq B_{\infty 1} j_1(j_1 + 1) \leq E^* \quad (2.15)$$

$$0 \leq B_{\infty 2} j_2(j_2 + 1) \leq E^* - B_{\infty 1} j_1(j_1 + 1) \quad (2.16)$$

$$|j_1 - j_2| \leq j \leq j_1 + j_2 \quad (2.17)$$

$$|J - j| \leq l \leq J + j \quad (2.18)$$

have to be obeyed. For the linear + linear molecule system, $W_l(E, J)$ now is given by the fourfold sum

$$W_l(E, J) = \sum_{j_1} \sum_{j_2} \sum_j \sum_l 1. \quad (2.19)$$

If the species 1 instead of a linear molecule is a spherical top, $\sum_{j_1}$ in Eq. (2.19) is replaced by $\sum_{j_1} (2j_1 + 1)$. If the other species 2 is also a symmetric top, $\sum_{j_2}$ in addition is replaced by $\sum_{j_2} (2j_2 + 1)$.

Systems with symmetric or asymmetric tops, instead of the spherical tops considered in sections 2.2 and 2.3, are treated in analogous manner. Instead of the $(2j_1 + 1)$ - or $(2j_2 + 1)$ -degeneracies, the relevant summations over the additional top quantum numbers are executed.

In principle, the summations of Eq. (2.19) can be elaborated in explicit way. However, in practice the treatment becomes untractable except in the limits of large J and $J = 0$. For these limits, replacing the sums by integrals, one obtains [9, 10] the results summarized in Table 1. In the following, for general values of J , we propose empirical approximate interpolations between these limiting expressions. Obviously, the fourfold summations of Eq. (2.19) can easily be evaluated numerically obeying the constraints (2.15)–(2.18). However, this requires some computer time, which becomes undesirable in large scale calculations, such that the use of a simpler empirical expression is justified.

3. Empirical Interpolation between Limiting Expressions for $W_l(E, J = 0)$ and $W_l(E, \text{high } J)$

In order to interpolate $W_l(E, J)$ between the limiting relationships for $W_l(E, J = 0)$ and $W_l(E, \text{high } J)$ which were derived in section 2, we use a doubly reduced representation. We reduce $W_l(E, J)$ by

$$W_l(E, J)/W_l(E, \text{high } J) = y_l, \quad (3.1)$$

and J by

$$(2J + 1) W_l(E, J = 0)/W_l(E, \text{high } J) = x_l. \quad (3.2)$$

The explicit quantum expressions of sections 2.1 and 2.2 for the atom + linear molecule and atom + spherical top systems obviously can be represented in doubly reduced form using y_l and x_l . After some algebraic transformations, for the atom + linear molecule system, one obtains

$$y_l = x_l - x_l^2/4 + 1/4 m^2 \quad (3.3)$$

with

$$m = j_{\max} + 1, \quad (3.4)$$

and

$$y_l = (1 - 1/12 m^2) x_l - n^2 x_l^3 / 12 m^4 \quad (3.5)$$

with

$$n = (2j_{\max}/3 + 1)(2j_{\max} + 1) \quad (3.6)$$

for the atom + spherical top system. With continuous variations of y_l and x_l , Eqs. (3.3) and (3.5) correspond to smooth functions, in contrast to the quantum expressions of sections 2.1 and 2.2.

For the more complicated systems of section 2.3, at first we have checked whether there exists a universal interpolation between the $x_l \rightarrow 0$ and $x_l \rightarrow \infty$ limits. In order to do this, for a variety of energies and combinations of rotational constants, we have numerically evaluated $W_l(E, J)$ following section 2.3. The systems treated are characterized in Table 2 and the results are illustrated in Fig. 1. The exactly counted values all were found in a very narrow band with

Table 1
Numbers of quantum states $W_l(E, J)$ for coupled systems of rotating species [9, 10] (for definition of symbols, see section 2)

System	$W_l(E, J = 0)$	$W_l(E, \text{high } J)$
atom + linear molecule	$E^{*1/2}/B_{\infty}^{1/2}$	E^*/B_{∞}
atom + spherical top	E^*/B_{∞}	$4 E^{*3/2}/3 B_{\infty}^{3/2}$
linear + linear molecule	$2 E^{*3/2} [B_{\infty 1}^{1/2} + B_{\infty 2}^{1/2} - (B_{\infty 1} + B_{\infty 2})^{1/2}]/3 B_{\infty 1} B_{\infty 2}$	$E^{*2}/2 B_{\infty 1} B_{\infty 2}$
linear (1) + spherical top (2)	$E^{*2} \arcsin [B_{\infty 1}^{1/2}/(B_{\infty 1} + B_{\infty 2})^{1/2}]/2 B_{\infty 1}^{3/2} B_{\infty 2}^{1/2}$	$8 E^{*5/2}/15 B_{\infty 1}^{3/2} B_{\infty 2}$
spherical + spherical top	$8 E^{*5/2}/15 B_{\infty 1} B_{\infty 2} (B_{\infty 1} + B_{\infty 2})^{1/2}$	$\pi E^{*3}/6 B_{\infty 1}^{3/2} B_{\infty 2}^{3/2}$

Table 2

Comparison of exactly counted numbers of states $W(E, J)$ with approximated values $W(E, J)_{\text{approx}}$ from Eqs. (4.1)–(4.3). (The largest J at each energy corresponds to the value at which W becomes independent of J)

E/cm^{-1}	J	$W(E, J)$	$W(E, J)_{\text{approx}}$
$\text{CH}_3 + \text{H}^{\text{a)}$			
1000	0	28.3	28.4
	2	138	137
	5	275	265
	8	351	337
	10	371	359
10000	0	3.17 E4	3.18 E4
	10	5.89 E5	5.87 E5
	20	8.53 E5	8.50 E5
	30	9.00 E5	8.98 E5
	35	9.01 E5	8.98 E5
$\text{C}_7\text{H}_7 + \text{H}^{\text{b)}$			
1000	0	9.16 E4	9.18 E4
	20	3.53 E6	3.54 E6
	40	5.92 E6	5.92 E6
	60	7.04 E6	7.05 E6
	80	7.36 E6	7.36 E6
10000	98	7.40 E6	7.38 E6
	0	7.82 E13	7.83 E13
	30	4.55 E15	4.58 E15
	60	7.95 E15	7.99 E15
	90	9.90 E15	9.99 E15
10000	150	1.10 E16	1.10 E16
	312	1.10 E16	1.10 E16
$\text{C}_7\text{H}_7 + \text{H}^{\text{c)}$			
1000	0	9.40 E4	9.12 E4
	20	3.63 E6	3.64 E6
	40	6.07 E6	6.07 E6
	60	7.22 E6	7.24 E6
	80	7.55 E6	7.55 E6
10000	98	7.59 E6	7.57 E6
	0	9.21 E13	9.24 E13
	30	5.36 E15	5.39 E15
	60	9.36 E15	9.42 E15
	90	1.17 E16	1.18 E16
10000	150	1.29 E16	1.30 E16
	312	1.30 E16	1.30 E16
$\text{HCO} + \text{H}^{\text{d)}$			
10000	0	6.23 E4	6.25 E4
	10	1.26 E6	1.26 E6
	20	2.20 E6	2.18 E6
	30	2.78 E6	2.73 E6
	40	3.01 E6	2.98 E6
30000	50	3.06 E6	3.05 E6
	53	3.06 E6	3.05 E6
	0	2.82 E6	2.82 E6
	10	5.83 E7	5.85 E7
	30	1.51 E8	1.50 E8
30000	60	2.17 E8	2.14 E8
	90	2.24 E8	2.23 E8
	93	2.24 E8	2.23 E8
$\text{CH}_3 + \text{OH}^{\text{e)}$			
5000	0	2.11 E5	2.11 E5
	5	2.20 E6	2.21 E6
	10	3.70 E6	3.68 E6
	15	4.53 E6	4.52 E6
	20	4.85 E6	4.84 E6
5000	25	4.92 E6	4.91 E6

Table 2 (Continued)

E/cm^{-1}	J	$W(E, J)$	$W(E, J)_{\text{approx}}$
10000	0	6.23 E6	6.24 E6
	5	6.63 E7	6.65 E7
	10	1.16 E8	1.16 E8
	15	1.51 E8	1.51 E8
	20	1.71 E8	1.71 E8
5000	25	1.80 E8	1.80 E8
	30	1.82 E8	1.82 E8
	35	1.83 E8	1.83 E8
$\text{OH} + \text{OH}^{\text{f)}$			
5000	0	1.05 E3	1.05 E3
	2	5.12 E3	5.17 E3
	4	8.82 E3	8.92 E3
	6	1.20 E4	1.21 E4
	8	1.44 E4	1.46 E4
10000	10	1.63 E4	1.65 E4
	15	1.91 E4	1.87 E4
	0	5.69 E3	5.70 E3
	5	5.93 E4	5.98 E4
	10	1.00 E5	1.01 E5
10000	15	1.24 E5	1.25 E5
	20	1.35 E5	1.36 E5
	22	1.39 E5	1.37 E5
$\text{CH}_3 + \text{CH}_3^{\text{g)}$			
5000	0	1.20 E6	1.20 E6
	5	1.26 E7	1.27 E7
	10	2.16 E7	2.16 E7
	15	2.72 E7	2.72 E7
	20	2.98 E7	2.98 E7
15000	25	3.06 E7	3.06 E7
	0	3.37 E9	3.37 E9
	5	3.63 E10	3.64 E10
	10	6.54 E10	6.56 E10
	15	8.83 E10	8.85 E10
15000	20	1.04 E11	1.05 E11
	30	1.20 E11	1.20 E11
	43	1.23 E11	1.23 E11
$\text{C}_2\text{H}_5 + \text{CH}_3^{\text{h)}$			
2000	0	9.34 E7	9.36 E7
	10	1.79 E9	1.79 E9
	20	2.77 E9	2.77 E9
	30	3.08 E9	3.08 E9
	40	3.11 E9	3.10 E9
10000	0	1.36 E13	1.36 E13
	20	4.83 E14	4.85 E14
	40	6.85 E14	6.90 E14
	60	7.19 E14	7.19 E14
	80	7.20 E14	7.18 E14
10000	90	7.20 E14	7.19 E14

Molecular Parameters:

^{a)} M. Quack and J. Troe, Ber. Bunsenges. Phys. Chem. 78, 240 (1974).

^{b)} D. C. Astholz, J. Durant, and J. Troe, 18th. Symp. (Int.) Combust., (Proc.), p. 885 (1980).

^{c)} As ^{b)} but with anharmonicity correction, see J. Troe, J. Chem. Phys. 66, 4758 (1977), individual threshold energy of each oscillator: 33000 cm^{-1} .

^{d)} J. Troe, J. Phys. Chem. 88, 4375 (1984).

^{e)} Methyl: see ^{a)}, hydroxyl, Ref. [5].

^{f)} Ethyl (with free internal rotor): W. L. Hase and H. B. Schlegel, J. Phys. Chem. 86, 3901 (1982).

a spread which did not exceed $\pm 3\%$. The limiting values given in Table 1 (i.e. $y_i \rightarrow 1$ for $x_i \rightarrow \infty$, and unit slope for $x_i \rightarrow 0$) were reproduced as well. However, for small energies, the best results in Fig. 1 were obtained, if exactly counted values were used for the reduction instead of the classical expressions given in Table 1. At small energies, this counting is easily done, whereas it becomes time consuming at large energy. As an example, we considered systems with rotational constants near 5 cm^{-1} ; at energies $E^* \geq 1000 \text{ cm}^{-1}$, the deviations between exactly counted and classical values (Table 1) for $W_i(E, J=0)$ and $W_i(E, \text{high } J)$ were smaller than 5%.

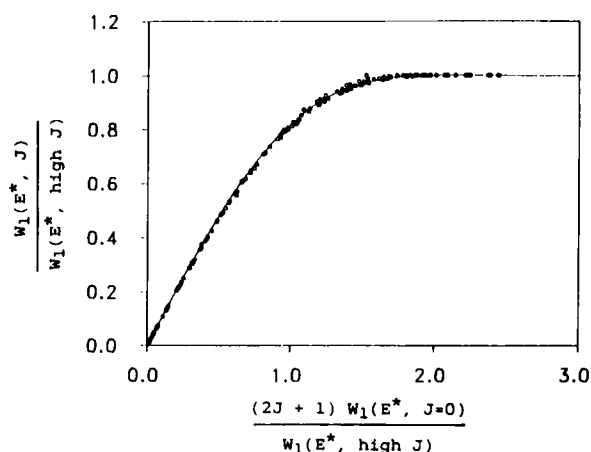


Fig. 1
Interpolation of number of quantum states $W_i(E^*, J)$ for freely rotating fragments in bond fission (points: accurate state counting for examples of Table 2, line: empirical interpolation from Eqs. (3.1), (3.2), and (3.7))

We have experimented with a number of empirical functions in order to represent the universal interpolation function of Fig. 1. We found that the function

$$y_i = \{0.08 x_i^{1/2} \exp[-1.1(x_i - 1)^2] + 1\} \tanh x_i \quad (3.7)$$

gives very good results. Fig. 1 demonstrates that Eq. (3.7) within $\pm 2.5\%$ agrees with the counted values.

Using the limiting classical expressions of Table 1 and Eq. (3.7), one obtains an empirical approximation for $W_i(E, J)$ at arbitrary J -values. Implementing this function for $W_i(E, J)$ into the starting array of a Beyer-Swinehart counting algorithm, applied to harmonic oscillators of the separated species, a particularly quick access to the total $W(E, J)$ is provided. If the separated species also contain free internal rotors, it appears most economic to convolute the limiting classical expressions for $W_i(E, J=0)$ and $W_i(E, \text{high } J)$ with the classical densities of states of the internal rotors by analogy to Eq. (1.1) and as described in Ref. 3. The J -interpolation according to Eq. (3.7) then should follow.

4. Empirical Interpolation between Limiting Expressions for $W(E, J=0)$ and $W(E, \text{high } J)$

An alternative to the procedure outlined in section 3 consists in the determination of limiting ($J=0$)- and (high J)-

expressions of the total number of states $W(E, J)$, and the subsequent interpolation between these limiting expressions for arbitrary J . By analogy to section 3 we have, at first, selected a number of molecular models and derived $W(E, J)$ by accurate count, including the oscillator states of the separated fragments and the rotational states of their coupled motion. The molecular systems considered were $\text{CH}_4 \rightarrow \text{CH}_3 + \text{H}$, $\text{C}_7\text{H}_8 \rightarrow \text{C}_7\text{H}_7 + \text{H}$, $\text{H}_2\text{CO} \rightarrow \text{HCO} + \text{H}$, $\text{CH}_3\text{OH} \rightarrow \text{CH}_3 + \text{OH}$, $\text{H}_2\text{O}_2 \rightarrow 2\text{OH}$, $\text{C}_2\text{H}_6 \rightarrow 2\text{CH}_3$, $\text{C}_3\text{H}_8 \rightarrow \text{C}_2\text{H}_5 + \text{CH}_3$. The energies were varied between 1000 and 50000 cm^{-1} above the dissociation threshold. Details of these models and the results are given in Table 2. A doubly reduced representation analogous to section 3 was chosen, using

$$W(E, J)/W(E, \text{high } J) = y \quad (4.1)$$

and

$$(2J + 1) W(E, J=0)/W(E, \text{high } J) = x. \quad (4.2)$$

Fig. 2 shows the results. Again, as in Fig. 1, all counted numbers of states fall into a very narrow band with a maximum spread of about $\pm 5\%$.

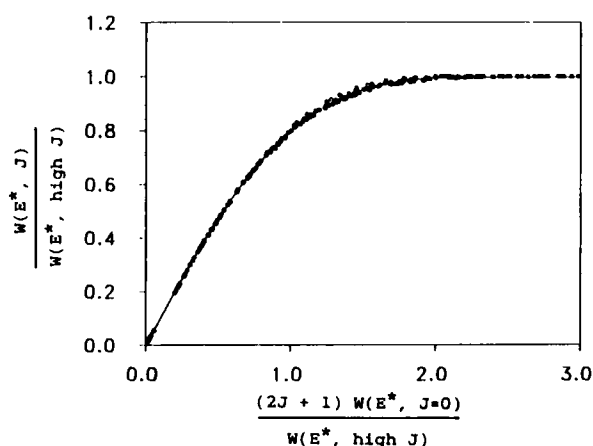


Fig. 2
Interpolation of total number of quantum states $W(E^*, J)$ of molecules with freely rotating fragments in bond fission (points: accurate state counting results of Table 2, line: empirical interpolation from Eqs. (4.1)–(4.3))

It turned out that the interpolation expression $y(x)$ was slightly below the graph of the expression $y_i(x_i)$ of Fig. 1. Fig. 2 includes an empirical representation of the results by the function

$$y = \{0.05 x^{1/2} \exp[-1.1(x - 1.15)^2] + 1\} \tanh x \quad (4.3)$$

The deviations between Eq. (4.3) and the exact values were always less than $\pm 5\%$. Larger deviations (up to $\pm 10\%$) were only obtained at energies near to that of the lowest vibrational quantum of the two fragments where a switch from $W_i(E, J)$ to $W(E, J)$ occurs. However, in this range a separate treatment for individual vibrational states is easily possible.

5. Interpolated Whitten-Rabinovitch Expressions for $W(E, J)$

For not too small energies, where exact counting would be easily performed, the Whitten-Rabinovitch [11, 12] approximation of densities and numbers of states has proved to be very useful in practice. Our present procedure has the advantage that it can easily be combined with the Whitten-Rabinovitch approach in order to obtain closed expressions of $W(E, J)$ for use in phase space theory or in adiabatic channel modified phase space theory. One adopts the treatment of section 4, i.e. one obtains the limiting ($J = 0$)- and (high J)-expressions first, and one then interpolates for general J using Eqs. (4.1)–(4.3). If the limiting classical expressions are abbreviated by

$$W_l(E) = A E^{*r/2}, \quad (5.1)$$

with the coefficients A and exponents r given in Table 1, on the Whitten-Rabinovitch level [12] the total numbers of states are given by

$$W(E) \approx A \frac{\Gamma(1+r/2) E^{*s+r/2}}{\Gamma(s+1+r/2) \prod_{i=1}^s \varepsilon_i}. \quad (5.2)$$

E^* now stands for

$$E^* = E - E_0(J) + a(E - E_0(J)) E_z \quad (5.3)$$

with the Whitten-Rabinovitch function $a(E')$, given by

$$a(E') = 1 - \beta w \quad (5.4)$$

with

$$\beta = \frac{(s-1)(s+r/2)}{s} \frac{\sum \varepsilon_i^2}{(\sum \varepsilon_i)^2} \quad (5.5)$$

and

$$\log w = -1.0506 (E'/E_z)^{0.25} \text{ at } E'/E_z \geq 1 \quad (5.6)$$

or

$$w^{-1} = 5(E'/E_z) + 2.73 \sqrt{E'/E_z} + 3.51 \text{ at } 0.1 \leq E'/E_z \leq 1. \quad (5.7)$$

$E_z = \sum_{i=1}^s \varepsilon_i/2$ is the zeropoint energy; the vibrational quanta ε_i correspond to the s vibrations of the separated fragments.

There will be one expression for $W(E, J = 0)$ and one for $W(E, \text{high } J)$, between which one interpolates using Eqs. (4.1)–(4.3). Free internal rotors of the separated fragments are easily included by increasing r and introducing the effective rotational constants and symmetry numbers [12].

The procedure outlined in this section presents the complete analogue to the interpolation described in section 4 where counted $W(E, J)$ were used. Examples for comparisons between counted $W(E, J)$ and the corresponding Whit-

ten-Rabinovitch results were given elsewhere [10–15] and need not to be repeated here. The additional convolution with fragment rotor states accounting for total angular momentum conservation, which is presented in the present work, proceeds exactly analogous to section 4 such that the accuracy of the method is preserved. When only very few fragment vibrational states contribute to $W(E, J)$, however, they should be treated with individual expressions for $W_l(E, J)$, see section 3, and the Whitten-Rabinovitch convolution should not be used.

6. Conclusion

The present empirical interpolation between ($J = 0$)- and (high J)-expressions of the numbers of quantum states in phase space theory considerably accelerates the calculation of specific rate constants $k(E, J)$ for isotropic fragment interaction potentials. It prepares a facile implementation of E - and J -dependent rigidity factors [16, 17] ($0 < f_{\text{rigid}}(E, J) < 1$) accounting for the anisotropy of the potential. It also liberates attention for the adequate calculation of anharmonic densities of states $\varrho(E, J)$ in the continuum energy range $E \geq E_0(J)$ such as they are relevant for characterizing excited state lifetimes.

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